Jatavath Bharath

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.21
Narayana Junior College	Class XII (Telangana Board of Intermediate Education)	2023	95.80%
Narayana High School	Class X (Board of Secondary Education, Telangana)	2021	10.00%

PROJECTS

Factor Models & Mean-Reversion Quant Engine

Python, NumPy, SciPy, Scikit-Learn, Matplotlib

- Built a statistical arbitrage research engine using 20 years of S&P 500 data (369 assets, 5,260 trading days); applied PCA to extract dominant market factors explaining ~70% of cumulative variance.
- Analyzed residuals using Marchenko-Pastur random matrix theory and modeled mean-reverting behavior with Ornstein-Uhlenbeck and AR(1) processes to estimate mean-reversion speed across different regimes.
- Conducted a 1,280-run parameter search to evaluate strategy configurations and achieved 10.7% return, 0.79 Sharpe ratio, and 11.3% maximum drawdown under 5 bps transaction costs.

Mobile PC Control Deck -- In Development

Python, Flask/FastAPI, HTML/CSS/JavaScript, Windows Automation

- Developing a mobile browser-based control interface that turns a smartphone into a customizable control deck for managing desktop applications and workflows remotely.
- Built a lightweight local server to establish real-time communication between the phone and Windows PC, enabling remote execution of system and application-level commands.
- Implemented controls for launching applications, switching virtual desktops, adjusting system volume/brightness, locking the PC, taking screenshots, and controlling trading-platform workflows.
- Currently expanding the system with additional automation, improved UI/UX, and reliable command handling for use across different desktop workflows.

Income & Wealth Distributions Using Kinetic Exchange Models

Python, NumPy, Matplotlib, Statistical Physics, Econophysics

- Developed Python simulations of kinetic exchange models to study how microscopic money-exchange and saving rules generate macroscopic wealth distributions across 1,000-10,000 agents and up to 500,000 trades per simulation.
- Implemented and analyzed three exchange models: no savings, uniform saving propensity, and heterogeneous saving propensities, reproducing exponential, Gamma-like, and Pareto-tailed wealth distributions respectively.
- Performed steady-state analysis to examine how saving behavior changes wealth concentration, with heterogeneous savings producing a power-law high-wealth tail ($P(m)$ m^2).

NinjaTrader Live Position Intelligence Tracker

TypeScript, Chrome Extension (MV3), WebSockets, Real-Time Event Processing

- Engineered a real-time browser extension to collect and process P&L and position-direction updates from multiple NinjaTrader traders simultaneously.
- Built an event-driven state engine to classify accounts as LONG, SHORT, or FLAT and continuously detect directional and P&L changes.
- Developed multi-account tracking with asynchronous background workers, maintaining synchronized trader states despite continuously changing leaderboard data.

- Created a real-time monitoring interface displaying P&L movement, trader position direction, rank changes, and account-level state transitions.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, HTML5/CSS3

AI & Data Systems: LLM Integration (OpenAI, Gemini), Retrieval-Augmented Generation (RAG), Vector Databases, Semantic Search

Web & Frameworks: React, Node.js, FastAPI, REST APIs, WebSockets

Tools & Infrastructure: Git, GitHub Actions, Docker, Linux/Bash, AWS (EC2, S3)

EXPERIENCE

Business Development Intern | ***Quikverse India Private Limited***

- Managed business operations and client outreach strategies to drive growth and meet performance targets.
- Collaborative cross-functional teams to execute market research and expand partnership opportunities.

JEE Mentor | **WE WON Academy**

- Mentored JEE aspirants, guiding students through college selection, admissions, and counselling.

- Supported students in evaluating college and branch options based on their ranks, preferences, and career goals.

POSITIONS OF RESPONSIBILITY

Logistics Coordinator, Sunshine Management Team

- Coordinated logistics and operational requirements for student events and team activities, working across multiple teams to ensure timely execution.
- Managed resource allocation, scheduling, and on-ground coordination while handling multiple requirements under tight deadlines.

EXTRACURRICULAR

NSO Hockey Player, IIT Hyderabad

- Represented IIT Hyderabad in institute-level hockey activities as part of the NSO program.
- Participated in regular team training and competitive activities, contributing to team coordination and consistent execution.